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Enhancing performance of vision transformers on small datasets through local inductive bias incorporation

Ibrahim Batuhan Akkaya [a,*], Senthilkumar S. Kathiresan [a], Elahe Arani [a,b,1], Bahram Zonooz [a,b,1]

[a] Advanced Research Lab, NavInfo Europe, Eindhoven, 5657 DB, Netherlands
[b] Department of Mathematics and Computer Science, Eindhoven University of Technology, Eindhoven, 5612 AZ, Netherlands

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ABSTRACT

Vision transformers (ViTs) achieve remarkable performance on large datasets, but tend to perform worse than convolutional neural networks (CNNs) when trained from scratch on smaller datasets, possibly due to a lack of local inductive bias in the architecture. Recent studies have therefore added locality to the architecture and demonstrated that it can help ViTs achieve performance comparable to CNNs in the small-size dataset regime. Existing methods, however, are architecture-specific or have higher computational and memory costs. Thus, we propose a module called *Local InFormation Enhancer (LIFE)* that extracts patch-level local information and incorporates it into the embeddings used in the self-attention block of ViTs. Our proposed module is memory and computation efficient, as well as flexible enough to process auxiliary tokens such as the classification and distillation tokens. Empirical results show that the addition of the LIFE module improves the performance of ViTs on small image classification datasets. We further demonstrate how the effect can be extended to downstream tasks, such as object detection and semantic segmentation. In addition, we introduce a new visualization method, Dense Attention Roll-Out, specifically designed for dense prediction tasks, allowing the generation of class-specific attention maps utilizing the attention maps of all tokens. The code for this project is available on Github (https://github.com/NeurAI-Lab/LIFEhttps://github.com/NeurAI-Lab/LIFE).

1. Introduction

Transformers, a new kind of encoder–decoder model that uses a self-attention mechanism to process input data [1], were initially proposed for sequence modeling in the domain of natural language processing (NLP). The success of transformers in NLP has led to the development of these architectures for a wide range of vision tasks [2–4]. Vision transformers (ViTs), when trained or pre-trained on a large dataset, outperform their CNN counterparts. However, for many real-world vision tasks, a large amount of annotated data is either too expensive or not feasible. As a result, the data-hungry nature of ViTs prevents them from being applied to a number of crucial real-world problems for which a limited amount of annotated data are available.

In the majority of ViTs, an image is divided into a sequence of non-overlapping patches, from which the self-attention layer learns the global context. Although spatially adjacent patches hold information for establishing local context, ViTs lack the inductive bias to leverage this due to limited encoding from a strided convolution in the patch embedding layer [2]. In contrast, convolutional layers in CNNs utilize pixel-level local information, enhancing data efficiency [5,6]. Leveraging local context might be pivotal for ViTs to effectively learn vision tasks with limited data samples.

Recent studies have improved the use of the local context in ViTs through architectural modifications [3], token pre-processing [7], or the addition of convolutional layers [8]. Although these findings support the importance of utilizing local information, their design is not adaptable enough to be annexed to other ViT architectures and/or negatively impacts other performance factors such as memory consumption and computational cost. Therefore, it is advantageous to devise a method to incorporate local information effectively and modularly into the architecture of ViTs.

The functionality of a transformer architecture depends on its self-attention layers, which require a sequence of embeddings. These embeddings are typically generated using a point-wise feedforward layer with a receptive field consisting of only one patch from the input image or one token from the previous layer (Fig. 1(a)). However, larger receptive fields can result in embeddings containing local information from spatially adjacent patches, as shown in Fig. 1(b). We hypothesize that

* Corresponding author.
E-mail address: bthakkaya@gmail.com (I.B. Akkaya).
[1] Equal contribution.

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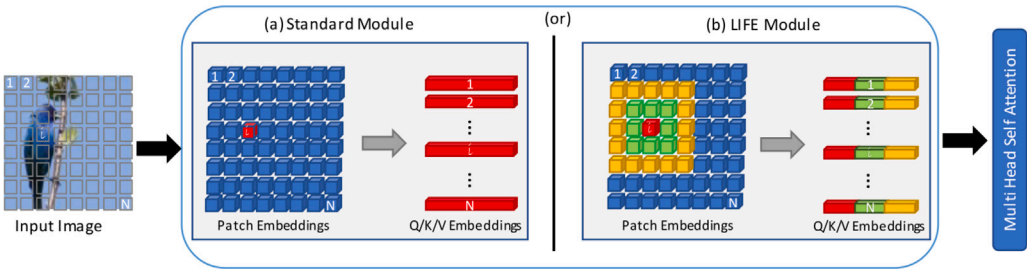

Fig. 1. In vision transformers, the input image is divided into N non-overlapping patches, transformed into embeddings, and processed through a point-wise feedforward layer to generate query Q, key K, and value V embeddings. These embeddings are then fed into a self-attention layer. Alternatively, LIFE module incorporates adjacent patch information into the Q, K, and V embeddings using increasingly larger receptive fields, expanding beyond the standard module's focus on individual patches (e.g., the ith $Q/K/V$ embeddings include information from adjacent patches, depicted in green and orange). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

by enriching the embeddings with this local context, vision transformer models (ViTs) will be able to learn the global context with fewer data points and perform better on smaller datasets.

The LIFE (*Local InFormation Enhancer*) module enhances vision transformer performance on smaller datasets by generating embeddings for self-attention layers with expanded receptive fields. By reshaping input tokens into an image format, employing convolution layers with various kernel sizes, and subsequently transforming feature maps back into tokens, the LIFE module introduces enhanced local context. This includes the use of depthwise separable convolutional (DSC) layers [9], specifically designed for auxiliary tokens such as classification and distillation tokens in ViTs, allowing more computationally efficient processing compared to standard conv layers.

We integrate the LIFE module into diverse ViT architectures of different capacities, evaluating its efficacy and versatility. Assessing classification performance on multiple datasets using DeiT, T2T, and Swin transformers as base architectures, we also explore its impact on dense prediction tasks with VOC and Cityscapes datasets for object detection and semantic segmentation. Our extensive empirical experiments consistently demonstrate LIFE's seamless integration across ViTs, enhancing performance across tasks with minimal memory and computation overhead. Additionally, we qualitatively analyze LIFE's contribution to each task, proposing a dense attention roll-out for visualizing dense prediction task attention. Our findings affirm LIFE's role in augmenting local context learning by guiding the network's attention to specific regions. Overall, our work underscores LIFE's adaptability, yielding substantial performance gains across diverse ViT architectures and tasks. Our contributions are as follows;

- Introducing *Local InFormation Enhancer (LIFE)* module complements global information by adding local context to the embeddings of ViT.
- Demonstration of the ability of the LIFE module to be easily integrated into different ViT architectures with minimal memory and computation costs overhead, even in the architecture that contains auxiliary tokens.
- Employing the LIFE module in different ViTs results in performance gains on smaller datasets such as ImageNet-100, Tiny-ImageNet, CIFAR10, and CIFAR100.
- LIFE module is versatile and also results in a boost in performance for dense prediction tasks.
- Proposing a novel method, *Dense Attention Roll-Out*, to visualize attention for dense prediction tasks.
- Qualitative evaluation of the contribution of the LIFE module to each task using our proposed visualization method.

2. Related work

Vision Transformers (ViT). have demonstrated competitiveness with convolutional neural networks (CNNs) in various vision tasks, such as image classification [3,10], object detection [11], semantic segmentation [12], and image inpainting [13]. These models utilize self-attention in the early stages to construct a convolution-free neural network. Following the introduction of the original ViT model [2], numerous studies have focused on improving classification performance and robustness through architectural modifications, knowledge distillation, or advanced data augmentation techniques [7,10,14,15].

Locality. Convolutional neural networks (CNNs) are a common choice for visual tasks, built with stacked convolution layers using small kernels for neighboring feature vectors. This design leverages spatial correlation in natural images, enhancing data efficiency. Evidence from CNNs suggests that it is essential to use local information to improve performance on small-scale datasets [5,6]. In contrast, the self-attention mechanism in the transformer block establishes a global relation between tokens, but ignores the locality. To address this, many approaches have been proposed to introduce locality bias into transformer architectures through architectural modifications.

Several recent studies have proposed hybrid networks to merge convolutional locality bias into transformer architectures. CeiT [16] uses low-level features from CNNs rather than patches extracted from raw images, and introduces a depth-wise separable convolution into the feedforward network within the transformer block to improve locality. CvT [17] integrates convolution into token embeddings, allowing a progressively decreasing number of tokens while increasing the dimension of the features. It also uses a depth-wise separable convolution to compute query, key, and value. Crossformer [18] addresses the lack of multiscale information in transformer architectures by processing tokens using short- and long-distance attention and combining multiscale information from neighboring tokens using multiple convolution layers with different kernel sizes within a pyramid-like architecture.

Another line of work proposes incorporating a pyramid structure derived from CNNs into the transformer architecture to induce locality in the network. This is achieved through various techniques, such as the use of a gradual shrinking technique in PVT [15], self-attention within windows in Swin [3,19], and hierarchical aggregation of transformer blocks in NesT [20]. All of these approaches involve the gradual combination of neighboring tokens, enabling the network to consider the local context in its processing.

Other approaches aim to introduce locality while maintaining the pure transformer architecture.

T2T [7] replaces the standard patch embedding layer with progressive tokenization to combine neighboring tokens; while TNT [21] divides the input image into large patches called visual sentences and small patches called visual words. TNT uses a shared sub-transformer architecture to extract the relation and similarity between the small patches, and passes the information generated by the sub-transformer to the visual sentence for further processing by the standard transformer block.

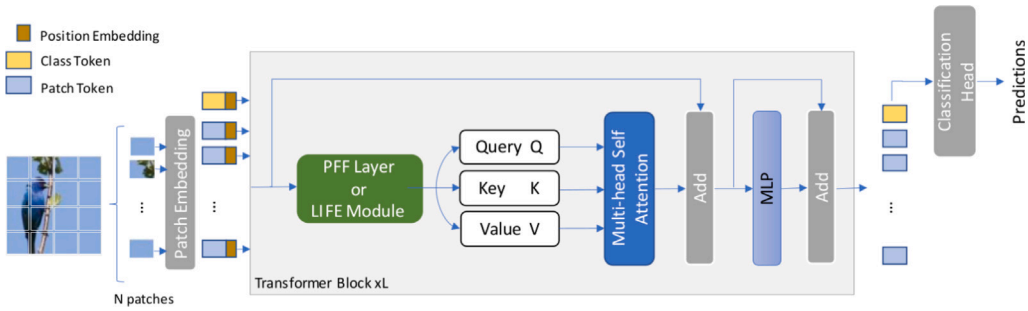

Fig. 2. Architecture of a standard ViT. To incorporate local context in embeddings for multi-head self-attention layers, we substitute the pointwise feedforward (PFF) layer with our novel LIFE module (details in Fig. 3), generating multiscale query, key, and value.

However, the mentioned methods are architecture-specific and focus on effective models for large datasets, potentially lacking performance on smaller ones. Liu et al. [22] propose a method that introduces a self-supervised task to predict the geometric distance between pairs of output tokens, while Li et al. [23] propose a distillation-based training mechanism that uses a lightweight pre-training CNN model to distill its features into a ViT. Although these approaches are flexible and can be integrated into different architectures, they may also come with additional memory and computational costs, and may not be as effective when applied to smaller datasets.

In contrast, the LIFE module is a modular and efficient approach that uses locality to generate richer embeddings, which is more effective in learning with less data. It introduces locality in the query, key, and value, making it flexible for integration with any transformer architecture. Unlike CvT, the LIFE module benefits from multiscale information and is more efficient compared to T2T and Crossformer, as it uses dense concatenation and depthwise separable convolution. Additionally, unlike previous work, the LIFE module utilizes multi-scale locality in every block and can be used in conjunction with training mechanisms for small-scale datasets.

3. Methodology

In ViTs, an input image is divided into N non-overlapping square patches [10]. These patches are then flattened and embedded in patch tokens X_{patch} of length C using a linear layer. Positional embeddings are added to the token patches and an additional classification token X_{cls}, which is a learnable embedding of the same length, forming a matrix X:

$$X = \left[X_{patch} | X_{cls}\right]; \qquad X \in \mathbb{R}^{C\times(N+1)}, X_{patch} \in \mathbb{R}^{C\times N}, X_{cls} \in \mathbb{R}^{C\times 1} \quad (1)$$

The resulting matrix is then passed to a series of L transformer blocks (Fig. 2), each consisting of a point-wise feedforward layer (PFF), followed by a multi-head self-attention layer (MHSA) and a multi-layer perceptron (MLP):

$$Q, K, V \leftarrow PFF(X); \qquad X_{out} = MLP(MHSA(Q, K, V)). \quad (2)$$

The final prediction is usually obtained by processing the classification token at the end of the last transformer block or by using the global average pooling of the patch tokens. However, standard ViTs lack local inductive bias; therefore, to introduce local context, we replace the PFF in Eq. (2) with our proposed LIFE module.

3.1. LIFE module

To incorporate local information, the LIFE module uses multiple hierarchically arranged convolutional layers, as depicted in Fig. 3. The first layer in the hierarchy is always a point-wise convolution with the smallest receptive field, while subsequent layers have progressively larger receptive fields due to increasing kernel sizes. The output features of these layers represent local information gleaned from various receptive fields, and are rearranged to obtain the final query Q, key K, and value V. The kernel sizes and paddings are configured to maintain constant spatial resolution throughout all layers, and for similar reasons, the channel size of all convolutional layer output feature maps is fixed at a constant value C'.

Except for the first layer, the LIFE module uses depth-wise separable convolutions [9] for all other layers, which consist of a depth-wise convolution followed by a point-wise convolution. This type of convolution is efficient in terms of memory and computation, allowing the LIFE module to have minimal overhead compared to the original PFF layer. In addition, point-wise convolution can be used to process any number of auxiliary tokens, such as the classification token and the distillation token. The LIFE module is described in more detail in the following.

Processing patch tokens in the LIFE module. The patch tokens $X_{patch} \in \mathbb{R}^{C\times N}$ are rearranged into an image format of shape $X_{p'} \in \mathbb{R}^{C\times H\times W}$, where $N = H \times W$. The $X_{p'}$ is then passed through a series of convolutional layers with progressively increasing receptive fields. Here, we use three layers:

$$F_{p_1} = Conv_1(X_{p'}); \qquad F_{p_2} = Conv_2(F_{p_1}); \qquad F_{p_3} = Conv_3(F_{p_2}); \quad (3)$$

where $F_{p_1}, F_{p_2}, F_{p_3} \in \mathbb{R}^{C\times H\times W}$ represent three different scales of local features obtained from the patch tokens. Each feature is divided into thirds along the channel dimension and reorganized into embeddings $Q_p, K_p, V_p \in \mathbb{R}^{C\times H\times W}$;

$$\begin{aligned} Q_p &= \left[F_{p_1}^{(C/3\times H\times W)}, \quad F_{p_2}^{(C/3\times H\times W)}, \quad F_{p_3}^{(C/3\times H\times W)}\right]^T; \\ K_p &= \left[F_{p_1}^{(C/3\times H\times W)}, \quad F_{p_2}^{(C/3\times H\times W)}, \quad F_{p_3}^{(C/3\times H\times W)}\right]^T; \\ V_p &= \left[F_{p_1}^{(C/3\times H\times W)}, \quad F_{p_2}^{(C/3\times H\times W)}, \quad F_{p_3}^{(C/3\times H\times W)}\right]^T. \end{aligned} \quad (4)$$

Processing auxiliary tokens in the proposed module. The class token $X_{cls} \in \mathbb{R}^{C\times 1}$ and any other auxiliary tokens can be processed using point-wise convolutions (PConv) in the LIFE module:

$$F_{c_1} = Conv_1(X_{cls}); \quad F_{c_2} = PConv_2(F_{c_1}); \quad F_{c_3} = PConv_3(F_{c_2}); \quad (5)$$

where $F_{c_1}, F_{c_2}, F_{c_3} \in \mathbb{R}^{C\times 1}$ represent different linear projections of the classification token for different scales. Next, each of the features is divided into three parts in the channel dimension and rearranged to form embeddings $Q_c, K_c, V_c \in \mathbb{R}^{C\times 1}$:

$$\begin{aligned} Q_c &= \left[F_{c_1}^{(C/3\times 1)}, \quad F_{c_2}^{(C/3\times 1)}, \quad F_{c_3}^{(C/3\times 1)}\right]^T; \\ K_c &= \left[F_{c_1}^{(C/3\times 1)}, \quad F_{c_2}^{(C/3\times 1)}, \quad F_{c_3}^{(C/3\times 1)}\right]^T; \\ V_c &= \left[F_{c_1}^{(C/3\times 1)}, \quad F_{c_2}^{(C/3\times 1)}, \quad F_{c_3}^{(C/3\times 1)}\right]^T. \end{aligned} \quad (6)$$

Subsequently, the features of the patch tokens $\in \mathbb{R}^{C\times H\times W}$ are flattened in spatial dimensions. The embeddings Q, K, and V from patch and auxiliary tokens concatenate into final embeddings $Q, K, V \in \mathbb{R}^{C\times(N+1)}$, which contain local features from different scales. The global information obtained through self-attention is complemented by the

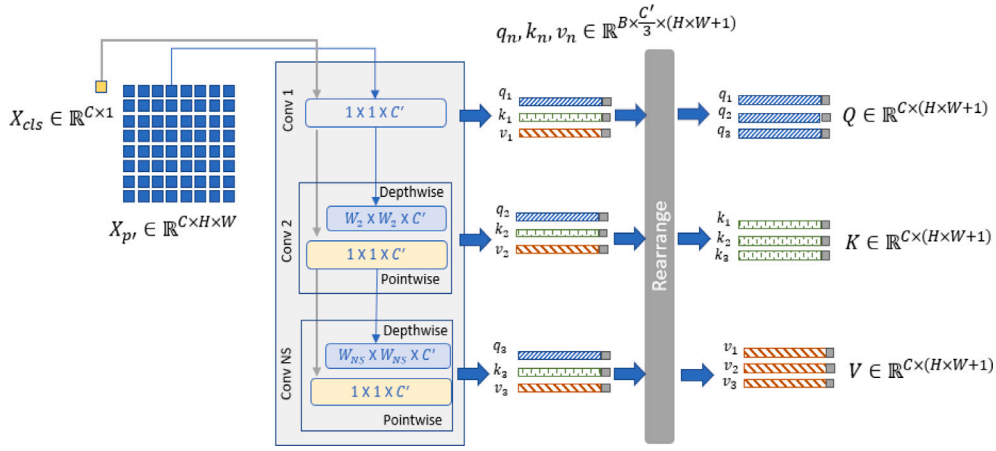

Fig. 3. Overview of *LIFE* module. It consists of a hierarchy of convolutional layers, with the first layer being a point-wise convolution and the remaining layers being depth-wise separable convolutions. Both the classification token X_{cls} and patch tokens $X_{p'}$ (arranged in the input image format) are passed through these layers (marked in ↓ and ↓, respectively). Each layer in the hierarchy outputs local information from a relatively larger receptive field. The information is then divided into query q_i, key k_i, and value v_i (including the processed class token represented as a gray box). Then, these hierarchical characteristics are rearranged to form the final query Q, key K, and value V. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

local context information encoded in these embeddings, resulting in improved performance. By combining global and local information, the model can better understand the context and relationships between different parts of the input.

4. Experimental settings

We analyze the LIFE module by integrating it into different state-of-the-art transformer architectures. The main experiments are conducted on small-scale image datasets, as our study focuses primarily on the performance of the transformer with limited data. We evaluate our model for the image classification task. However, many real-world applications are based on object detection or semantic segmentation, so we also examine how the module affects downstream dense prediction tasks. We demonstrate the effectiveness of our module for small datasets with quantitative and qualitative results in classification, detection, and segmentation tasks.

Datasets. Small-scale datasets used in our experiments include CIFAR-100 [24], CIFAR-10 [24], TinyImageNet [25], and ImageNet-100 [26]. These datasets contain 50k, 60k, 100k, and 130k training samples, respectively. ImageNet-100 and TinyImageNet are subsets of the ImageNet-1k dataset [26], with 200 and 100 classes, respectively. CIFAR-10 and CIFAR-100 have 10 and 100 classes, respectively. Except for the ImageNet-100 datasets, all other datasets have small image resolutions, either 32 × 32 or 64 × 64.

For experiments on extremely small classification datasets, we used FGVC-Aircraft [27], Oxford 102 Flower [28], Oxford-IIIT Pet [29], and Stanford Cars [30] datasets. These contain 6667, 2040, 3680, and 8144 samples in their training sets, with 3333, 6149, 3669, and 8041 samples in their test sets. They encompass 100, 102, 37, and 196 classes, respectively.

For object detection, VOC dataset [31] is used, and the Cityscapes dataset [32] is used for semantic segmentation tasks. The VOC and Cityscapes datasets contain 1464 and 5000 samples for training, respectively.

Implementation details. In our experiments, we used the Tiny and Small variants of the DeiT [10], T2T-ViT-12, and Swin transformers as baseline models for the classification task. We obtained the LIFE variants of these models by replacing the linear projection that generates the query, key, and value with the LIFE module. We employ three scales with kernel sizes of 1, 3, and 5 and zero paddings of sizes of 0, 1, and 2, respectively.

We used the original image size in our experiments. For the Swin transformer, we select a window size of 8, and for T2T-ViT-12, we decrease the token dimension from 255 to 252 in order to process the token in a multiscale manner in the LIFE module for all datasets. Baselines were designed for an input image size of 224 × 224. Therefore, we keep the network configurations the same as the baselines for the ImageNet-100 dataset. Only for the Swin architecture, we resize the input to 256 × 256. For other datasets with small image sizes, we used a patch size of 1 for Swin and 4 for DeiT architectures. To adapt T2T-ViT-12, we replaced the first unfold operation with a 3 × 3 kernel with a stride of 2 and the last unfold operation with a 1 × 1 kernel.

For detection and segmentation tasks, we use a tiny version of DeiT and Swin as the backbone. In order to highlight the effect of the LIFE module, we employ simple heads for dense prediction tasks. For the object detection task, we exploit DETR [11] and evaluate different combinations of backbone and head with and without the LIFE module. For segmentation tasks, we use simple upsampling after the linear transformation operation as the segmentation head. The input image size for both tasks is 512 × 512.

For the classification, detection, and segmentation tasks, we followed the training details in [10,33] and [12]. We re-train all models from scratch with random initialization in our framework for a fair comparison. We use a batch size of 512 for all datasets. For the dense prediction tasks, we also present results for ImageNet pre-trained initialization, as it is believed that the initialization can significantly impact the final performance.

Unlike transformer models, we observed that ResNet [5] benefited more from simple augmentations on small datasets. Thus, we did not apply any augmentation other than random crop and random horizontal flip. ResNet models are trained with SGD with a momentum of 0.9 for 200 epochs. The initial learning rate is set to 0.1 and adjusted with a multi-stage scheduler, which multiplies the learning rate by 0.2 at epochs 60, 120, and 160.

5. Quantitative results

In the following, we present the results of the evaluation of the LIFE module on various tasks and datasets. We first examine the efficacy of the LIFE module in addressing the performance gap between the transformer and the CNN counterpart when trained on smaller image classification datasets. We then evaluate the versatility of the LIFE module by employing it for dense prediction tasks, including object detection and semantic segmentation.

Table 1
Performance comparison of ViTs on small-scale image classification datasets with and without the addition of the LIFE module. The results include Top-1 accuracy, number of parameters, and GMAC for DeiT and T2T architectures with an input size of 224 × 224 × 3, and for the Swin architecture with an input size of 256 × 256 × 3.

		#Params	GMAC	CF-10	CF-100	Tiny-IM	IM-100
CNN	Resnet-18	11.23	2.37	94.14	75.10	60.86	80.74
	Resnet-50	23.71	5.35	94.32	74.25	63.45	82.70
Transformers	DeiT-T	5.54	1.26	85.65	51.45	54.89	63.56
	DeiT-T-LIFE	5.62	1.27	**89.28**	**71.74**	**60.51**	**68.32**
	T2T-ViT-12	6.66	1.74	88.41	52.51	57.89	83.58
	T2T-ViT-12-LIFE	6.59	1.71	**89.96**	**54.51**	**60.52**	**83.96**
	DeiT-S	21.70	4.61	87.24	70.39	55.08	80.84
	DeiT-S-LIFE	21.85	4.64	**90.38**	**73.97**	**59.78**	**81.52**
	Swinv2-T	27.72	5.95	95.03	76.65	64.78	86.86
	Swinv2-T-LIFE	27.88	6.01	**95.14**	**77.48**	**65.84**	**86.98**

5.1. Image classification

We assess the efficacy of the LIFE module in bridging the performance gap between the transformer and the CNN counterpart on limited datasets. The LIFE module aims to mitigate this by infusing locality bias into the transformer architecture via convolutional layers. We train models on a compact image classification dataset and contrast their performance with equivalent-sized CNNs like ResNet. As depicted in Table 1, the efficiency of LIFE module is emphasized by its integration into various models of various sizes and architectures. Accuracy serves as a performance metric for comparison. Results reveal that incorporating the LIFE module into the transformer architecture enhances performance on modest datasets across varied architectures and model sizes. This enhancement is more notable for smaller model sizes; for instance, the LIFE module boosts the DeiT-Tiny architecture's performance by about 15% and the DeiT-Small architecture by around 5% on average on the small dataset.

Although the T2T-ViT-12 and Swin transformers already incorporate some degree of locality through their modified architectural designs (i.e., the token-to-token module in T2T-ViT-12 processes overlapping windows; and the Swin has a pyramid architecture that combines neighboring tokens at each stage), the integration of the LIFE module further improves performance by providing multi-scale locality in every block. Overall, the improvement gain is more significant for DeiT, which lacks local information in its architecture, compared to the T2T and Swin Transformers.

We also report the number of parameters and GMACs required to infer an image size of 224 × 224 for T2T and DeiT, and 256 × 256 for the Swin transformer. The LIFE module minimally affects parameters and computations. Notably, T2T-ViT-12-LIFE is more efficient than the baseline, with an embedding size of 252 vs. 256 for T2T-ViT-12.

5.1.1. Classification on extremely small datasets

To showcase LIFE's adaptability to extremely small datasets, we conducted additional experiments across various benchmarks, encompassing the Stanford Cars, FGVC-Aircraft, Oxford 102 Flower, and Oxford-IIIT Pet datasets. In these experiments, we employed a commonly utilized transfer learning strategy by initializing models with pre-trained weights from the ImageNet dataset. We maintained consistency in our hyperparameter configuration as applied in our image classification studies, with a few deviations. Specifically, we reduced the number of training epochs to 100, excluded repetitive augmentation, and lowered the weight decay rate to 1e−8.

Table 2 demonstrates that, consistent with the image classification framework, LIFE prominently improves performance across diverse architectures and limited datasets. In most instances, the integration of the LIFE module yields performance improvements; in a few cases, it achieves accuracy comparable to the baseline model. Particularly noteworthy are the results on the Flowers and Pets datasets, where the DeiT-T model encountered convergence issues. Through the incorporation of the LIFE module, the model achieved a satisfactory level of accuracy, which is significant given the limited training samples from the datasets. This finding underscores the increasing importance of locality as sample availability diminishes and the effective mitigation of this challenge by the inclusion of the LIFE module.

Table 2
Performance comparison of ViTs on extremely small-scale image classification datasets with and without the addition of the LIFE module. Results exhibit top-1 accuracy at 224 × 224 × 3 input size, and for Swin at 256 × 256 × 3.

	Aircraft	Flowers	Pets	Cars
DeiT-T	73.84	35.60	25.62	83.97
DeiT-T-LIFE	**75.34**	**91.67**	**90.65**	**84.40**
T2T-ViT-12	82.33	96.57	92.67	**90.83**
T2T-ViT-12-LIFE	**83.14**	**97.32**	**92.83**	90.71
DeiT-S	80.05	**97.14**	**93.05**	91.08
DeiT-S-LIFE	**81.49**	97.02	**93.05**	**91.59**
Swinv2-T	85.51	98.21	93.68	92.23
Swinv2-T-LIFE	**85.81**	**98.54**	**93.95**	**92.31**

Table 3
Performance comparison on CIFAR-100: Locality-Free (DeiT-T), One-Scale (DeiT-T-OneScale), and Multi-Scale (DeiT-T-LIFE) Models.

Model	Deit-T	Deit-T-OneScale	Deit-T-LIFE
Accuracy	51.45	60.64	**71.74**

Overall, these results demonstrate that the addition of the LIFE module improves the performance of ViTs on various small datasets. This effect is more prominent when the model is smaller and the dataset is more complex (with a higher number of classes and fewer samples per class). Additionally, the LIFE module can be easily integrated into different architectures with minimal memory and computational overhead.

5.2. Impact and importance of multiscale embeddings

To demonstrate the effectiveness of multiscale information, we considered a modified version of the LIFE module called LIFE-OneScale, which encodes information using a single scale with a kernel size of three, similar to CvT. We evaluated this configuration by integrating it into DeiT-Tiny on the CIFAR-100 dataset (Table 3). Using a single scale, we observed over 9% improvement over the baseline. However, encoding multiscale information with three kernels of sizes 1, 3, and 5 resulted in over 20% improvement over the baseline. These results emphasize the advantage of using multi-scale information.

5.3. Object detection

For object detection tasks, we use the DETR architecture [11], which consists of a feature extractor as the backbone and a transformer

Table 4
Evaluation of the effectiveness of adding the LIFE module to ViTs in the object detection task using the DETR architecture with DeiT-T as the backbone, trained and tested on the VOC dataset.

	Backbone	Head	# params	GMAC	mAP	F1-Score
Random Initialization	DeiT-T	EncDec	27.20	14.20	27.31	37.52
		EncDec-Linear	28.38	14.50	29.76	40.24
		EncDec-LIFE	28.19	14.45	**31.37**	**41.80**
	DeiT-T-LIFE	EncDec	27.27	14.28	27.09	37.49
		EncDec-Linear	28.46	14.58	29.22	39.60
		EncDec-LIFE	28.26	14.53	**32.21**	**42.52**
ImageNet-1k Initialization	DeiT-T	EncDec	27.20	14.20	72.12	79.32
		EncDec-Linear	28.38	14.50	73.02	80.12
		EncDec-LIFE	28.19	14.45	**73.51**	**80.42**
	DeiT-T-LIFE	EncDec	27.27	14.28	72.43	79.68
		EncDec-Linear	28.46	14.58	73.05	80.16
		EncDec-LIFE	28.26	14.53	**73.92**	**80.73**
	Swinv2-T	EncDec	50.07	27.32	78.21	84.48
		EncDec-Linear	51.26	27.62	78.94	85.01
		EncDec-LIFE	51.06	27.57	**80.57**	**86.13**
	Swinv2-T-LIFE	EncDec	50.23	27.57	80.24	85.92
		EncDec-Linear	51.41	27.88	**81.27**	**86.73**
		EncDec-LIFE	51.22	27.83	**81.27**	86.68

Table 5
Evaluation of the effectiveness of adding the LIFE module to ViTs on the semantic segmentation task using the DeiT-T architecture as the backbone, trained and tested on the Cityscapes dataset. The results include the mean IoU (mIoU), the number of parameters, and the computation cost (in GMAC).

	Model	# params	GMAC	mIoU
Random Initialization	DeiT-T	10.13	10.50	31.89
	DeiT-T-LIFE	10.21	10.58	**33.73**
ImageNet-1k Initialization	DeiT-T	10.13	10.50	58.62
	DeiT-T-LIFE	10.21	10.58	**59.17**
	Swinv2-T	32.36	23.95	59.37
	Swinv2-T-LIFE	32.52	24.20	**60.25**

encoder–decoder (EncDec). In the encoder, the features extracted by the backbone are flattened and used as query Q, key K, and value V in the self-attention layer of the encoder. In the decoder, the encoder output is used as Q and K. V is defined as a learnable parameter that is later used to predict the final details of the object.

To evaluate the effect of the LIFE module, we integrate it into both the backbone and the encoder–decoder. The DETR encoder–decoder does not include a linear transformation to generate Q, K, and V. To ensure a fair comparison, we first add a linear layer to the encoder–decoder just before self-attention to generate Q, K, and V from the backbone features, which is denoted as EncDec-Linear. Then, we replace this linear layer with the LIFE module, referred to as EncDec-LIFE. We use the encoder–decoder without any transformation as a baseline, which is referred to as EncDec. We use the DeiT-T and Swinv2-T architectures and their LIFE variants as the backbone.

Table 4 shows the mAP and F1 scores for different combinations of the backbone and encoder–decoder. We observe that adding linear layers to generate Q, K, and V leads to improvement. Replacing the linear layer with the LIFE module leads to additional improvement, indicating that the local information from neighboring patches can aid in more accurate object detection in a scene. The best results are obtained when the LIFE module is used in both the backbone and the encoder–decoder.

We also evaluate the performance of the LIFE module with two different initializations. When models are randomly initialized, the LIFE module improves performance by ∼+4 mAP. When ImageNet-1k pre-trained weights are used for initialization, the LIFE module improves performance by ∼+2 mAP. These results suggest that the inclusion of locality bias can be beneficial when training a model from scratch with a small dataset.

5.4. Semantic segmentation

For the semantic segmentation task, we use a simple segmentation architecture to assess the efficacy of the LIFE module. We utilize DeiT-T and Swinv2-T architectures as the backbone, with a linear layer and upscaling for the segmentation head to align feature dimensions with class count. DeiT-T-LIFE and Swinv2-T-LIFE denote models with integrated LIFE modules across all backbone attention layers.

Table 5 shows the performance in terms of the number of parameters, GMAC, and mIoU results for both models and two initialization methods. The LIFE module improves performance in both cases. When the model is initialized with random weights, the LIFE module improves performance by ∼2%. If ImageNet-1k pre-trained weights are used for initialization, the improvement is 0.55%. Similar to the object detection task, we observe a greater improvement when the model is trained from scratch.

6. Qualitative results

To understand the decision-making process of a transformer model, we present the results of qualitative analyses using attention maps derived from transformer architectures. First, we generate attention visualizations for the classification task employing the existing *Attention Roll-Out* method [35]. Then, we propose the *Dense Roll-Out* method, which generates class-specific attention maps for dense prediction tasks, and demonstrate the effectiveness of the LIFE module utilizing our proposed visualization method.

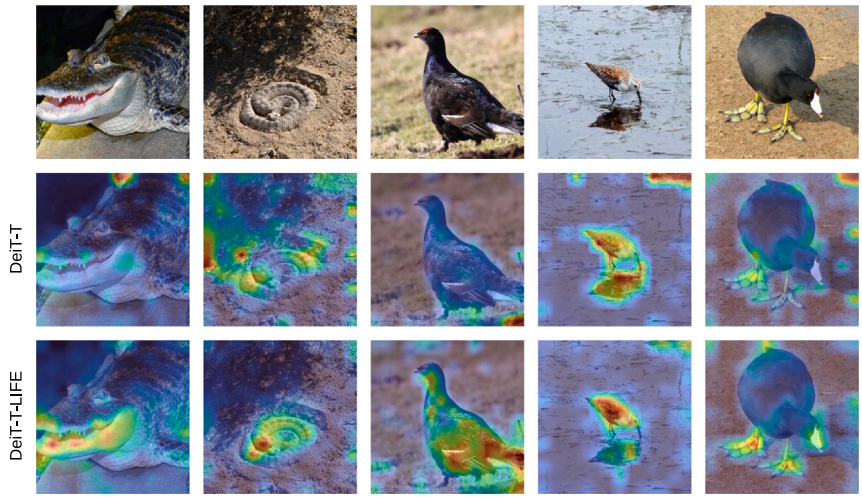

Fig. 4. Comparison of DeiT-T attention maps with and without the LIFE module, trained on the ImageNet-100 dataset, using the Attention Roll-Out method [34].

6.1. Image classification

We present a qualitative analysis of attention maps generated from DeiT-S with and without the LIFE module, trained on the ImageNet-100 dataset. As shown in Fig. 4, the attention of a standard DeiT-T is scattered between the background and the foreground. However, when the LIFE module is used, the ViT architecture gains more local information, enabling it to better identify the foreground object and focus more on it.

6.2. Dense prediction tasks

Dense attention roll-out. Attention mechanisms in transformer architectures allow the model to consider contextual information about the relationships between input tokens within a transformer block. Attention visualizations, which display the attention weights assigned to different input tokens, can provide insight into how the model makes its decisions and how well it can generalize to unseen data. There have been several methods proposed in the literature for generating attention visualizations in classification tasks, such as Attention Roll-Out [35] and Gradient Attention Roll-Out [36]. These methods have been effective for classification tasks in vision transformers, but there is currently no method proposed for dense prediction tasks, such as object detection and semantic segmentation.

To fill this gap, we present a novel approach, *Dense Attention Roll-Out*, to generate attention visualizations in dense prediction tasks. This method is based on the Attention Roll-Out method, which creates a pairwise attention graph by linearly combining attention from all blocks in a transformer architecture. However, we modified the method to specifically target dense prediction tasks. Unlike the standard method using only the attention map corresponding to the classification token, our method employs attention maps of all tokens, as all patch tokens contribute to dense prediction tasks. Moreover, we use simple heads to generate attention using backbone features, where most of the relevant information for prediction is located.

To generate class-specific attention maps, we align network predictions spatially with relevant tokens for predicting a specific class, employing either a segmentation map or a bounding box based on the task. Averaging the attention maps of these tokens and eliminating the global content yields the final class-specific attention map. Global content is common information to all tokens, derived by averaging all tokens.

Table 6
Inference and training time overhead of the LIFE module.

Model	GMACs	Forward time (ms)	Backward time (ms)
Swinv2-T	27.72	25.52	32.08
Swinv2-T-LIFE	27.88	32.50	47.07

Result. To illustrate the impact of the LIFE module, we opted for straightforward heads in both object detection and segmentation tasks. Utilizing simple heads enables visualization using backbone features, where crucial prediction information resides. Figs. 5 and 6 compare models with and without the LIFE module integrated into the backbone for object detection and segmentation tasks, respectively. The visuals present input images, predictions, and global attention in the first three columns. Subsequent columns display class-specific attention maps, featuring class names at the top. These attention maps demonstrate that the LIFE module enhances the model's focus on relevant input image regions, resulting in more precise predictions.

7. Limitations and future work

Our study demonstrates the efficacy of incorporating the LIFE module, resulting in improved classification accuracy within pretrained scenarios. However, a substantial challenge persists: achieving convergence during scratch training for both baseline and LIFE-enhanced models. This challenge holds particular significance for broader machine learning applicability, especially in scenarios with limited available data.

Additionally, LIFE module utilizes successive convolutions for multiscale information extraction involving token preparation, convolution output management, and various operations such as reshaping, indexing, and concatenation. While these operations marginally increase the GMACs cost, they impose significant demands on memory access, potentially leading to increased memory allocation and deallocation. This could consequently affect overall performance. For instance, a detailed assessment of the training time for the Swinv2-T model, conducted on a Tesla T4, is presented in Table 6.

It is pertinent to note that our primary focus did not center on optimizing the deployment of the LIFE module. Therefore, the study did not extensively consider the implementation's optimization. Nevertheless, an improved implementation holds promise for significantly reducing inference time, opening avenues for future exploration and enhancement in real-time applications.

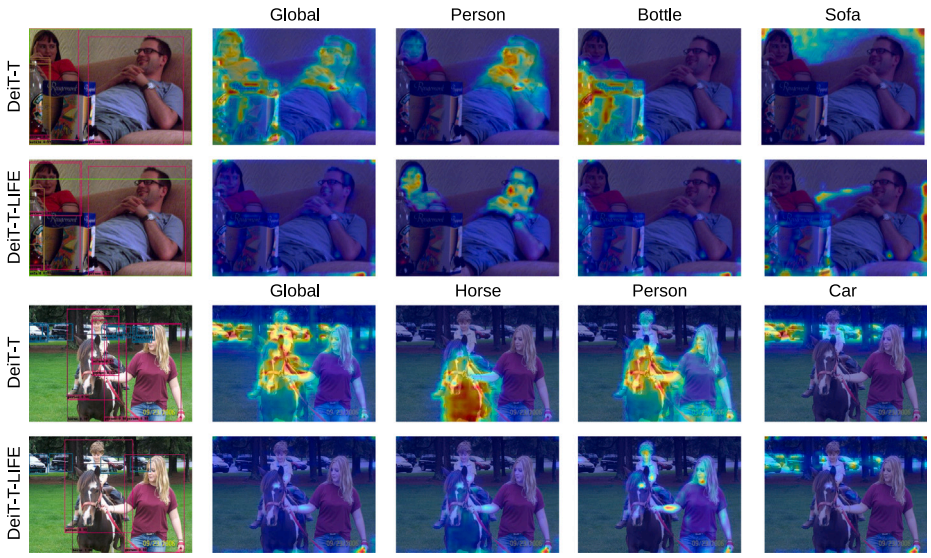

Fig. 5. Comparison of attention maps of object detection models with and without the LIFE module using the proposed Dense Roll-Out method.

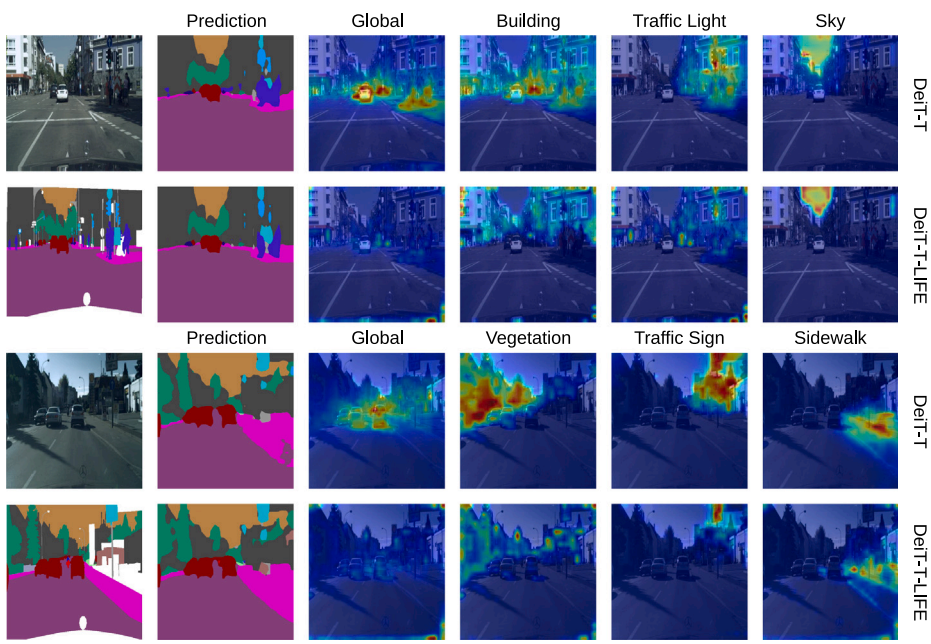

Fig. 6. Comparison of attention maps of segmentation models with and without the LIFE module using the proposed Dense Roll-Out method.

8. Conclusion

We proposed a novel module, *Local InFormation Enhancer (LIFE)*, for vision transformers (ViTs) that effectively leverages local information from images to generate more informative embeddings for the self-attention layers in each transformer block. We evaluated the impact of the LIFE module on models with different sizes (DeiT-Tiny and DeiT-Small) and architectures (DeiT, T2T, and Swin). The classification results showed that our proposed method consistently improved performance on small datasets.

Moreover, we observed more substantial improvement with lower-capacity models, indicating the LIFE module effectively leverages the introduced locality bias. It also proved effective within the hierarchical transformer architecture, inherently capitalizing on multiscale information, underscoring the importance of locality even at the window level.

Additionally, integrating LIFE module into ViTs for object detection and segmentation tasks notably enhanced performance, showcasing its adaptability and efficacy across diverse tasks. We noted that initializing the model with parameters learned from the ImageNet dataset yielded superior improvements compared to training from scratch. This underscores how the local information encoded by the LIFE module aids ViTs in achieving enhanced performance across vision applications like classification, object detection, and segmentation, with minimal computational or memory overhead.

Finally, we introduced 'Dense Attention Roll-Out', a novel visualization approach designed for dense prediction tasks like object detection and semantic segmentation. This method generates class-specific attention maps from all tokens' attention maps, offering insights into the decision-making process and the transformer architecture's generalization capabilities.

CRediT authorship contribution statement

Ibrahim Batuhan Akkaya: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Senthilkumar S. Kathiresan:** Formal analysis, Software, Writing – review & editing. **Elahe Arani:** Project administration, Supervision, Writing – review & editing. **Bahram Zonooz:** Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The study employs datasets that are publicly available.

Appendix. Comparison with prior works

Several architectures, such as ConVit, Crossformer, and CvT integrate locality. We aim to highlight LIFE's novelty compared to these architectures.

ConViT employs a gated positional self-attention mechanism (GPSA) that integrates positional information into a "soft" convolutional inductive bias. It enhances the self-attention block by including positional details. Conversely, LIFE modifies the query, key, and value embeddings, integrating multiple convolutional operations to infuse multiscale information into these embeddings prior to the self-attention process.

Crossformer proposes two modules, Cross-scale Embedding Layer (CEL) and Long-Short Distance Attention (LSDA) modules, to incorporate locality into their design. The CEL module integrates various patches with different scales into each embedding to provide cross-scale features to the self-attention module. The LSDA module divides the self-attention module into short-distance and long-distance components to reduce computational load while maintaining small- and large-scale features in the embeddings. Our LIFE module shares an objective similar to the CEL module, as both leverage multiscale information. However, the LIFE module offers a more efficient and adaptable design compared to the CEL module. Specifically, the LIFE module utilizes depthwise separable convolution, which provides superior efficiency compared to the CEL module. Additionally, the LIFE module can handle auxiliary tokens, improving its flexibility for integration with diverse architectures. In contrast, the Crossformer design does not incorporate auxiliary tokens, which prevents the CEL module from having this functionality.

The introduction of locality in CvT has been accomplished through two mechanisms: a convolutional patch embedding layer and a convolutional query, key, and value projection. The convolutional projection layer in CvT is similar to our own work; however, CvT employs only a single scale with a kernel size of three. In contrast, our LIFE module encodes multiscale information utilizing three scales with kernel sizes of 1, 3, and 5.

To perform a comparative analysis of the efficacy of the LIFE module with state-of-the-art approaches, we evaluated the performance of the CvT-21 model, which has 30 million parameters, on the CIFAR-100 dataset. Our results indicate that CvT-21 achieves an accuracy of 76.07%. The Swin Transformer with the LIFE module achieved an accuracy of 77.48%. This finding suggests that the LIFE module, in combination with a strong backbone, can outperform the CvT-21 model even with fewer parameters.

It is noteworthy that our study aims to imbue locality into the transformer framework with modularity and efficiency. To attain this objective, we introduced the LIFE module to efficiently utilize multi-scale data and adapt across architectures. The successful integration of the LIFE module was showcased in various designs, encompassing the standard transformer (Deit), pyramid architecture (Swin), and a custom patch embedding (T2T). Our results demonstrate that integrating the LIFE module effectively incorporates local information, enhancing performance on limited datasets.

The locality bias enhances performance with limited training data by focusing on crucial local information for effective learning. While advantageous in low-data scenarios, in large datasets where ViTs excel at capturing global context and long-range dependencies, the local emphasis of the LIFE module may yield diminishing returns as the dataset size increases.

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Ibrahim Batuhan Akkaya is a deep learning researcher. He holds a Ph.D. in Electrical and Electronics Engineering from Middle East Technical University. He worked in the defense industry for several years. His research interests are computer vision, vision transformers, and transfer learning.

Senthilkumar Sockalingam Kathiresan is a deep learning research engineer with 3 years of experience in the computer vision and deep learning industry. His passion lies in optimizing deep neural network architectures for improved accuracy and real-world deployment.

Elahe Arani, Ph.D. in computational neuroscience, is AI Assistant Professor at Eindhoven University of Technology. She develops innovative AI models inspired by human brain mechanisms and has published over 40 papers in top AI, computer vision, and neuroscience journals and conferences.

Bahram Zonooz is an accomplished executive with a strong background in autonomous vehicles and mapping. He is also an AI professor at Eindhoven University of Technology, where his team is at the forefront of developing neuro-inspired AI models for efficient learning, publishing in top-tier venues such as ICML, ICLR, and AAAI.